Madeleine S. Gastonguay

Biomedical Engineering and Institute for Computational Medicine at Johns Hopkins University

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PhD candidate in Biomedical Engineering (focus area: Computational Medicine) with expertise in mechanistic modeling, viral and immune dynamics, QSP, and Bayesian inference. Skilled in building ODE, stochastic, and nonlinear mixed-effects models to gain mechanistic insights into biological processes and support therapeutic design. Pursuing a career applying mathematical and statistical modeling to drug development and clinical translation for novel therapeutic modalities.

Skills & Certifications

Modeling: ODE, DDE, stochastic modeling, QSP, PK/PD, PBPK, NLME, structural and practical identifiability analysis, sensitivity analysis and parameter optimization, Bayesian inference
Programming: R (*tidyverse*, *deSolve*, *Shiny*, *plotly, brms*), Monolix (*lixoftConnectors*), MATLAB, Python, Bash, HPC/SLURM, SQL, Stan, JAGS
Teaching: Certified Instructor with The Carpentries

Education

Johns Hopkins School of Medicine, Baltimore, MD — Aug. 2022 – *May 2027 (expected)*
PhD in Biomedical Engineering
Institute for Computational Medicine
GPA: 4.0/4.0
Advisors: Dr. Alison L. Hill & Dr. Feilim Mac Gabhann

University of Connecticut, Storrs, CT — Aug. 2016 – May 2020
Bachelor of Science, Applied Mathematics
Summa Cum Laude with Honors; GPA: 3.98/4.00
Minor: Bioinformatics
Thesis: A Quantitative Pipeline for the Identification of Combinations of Targets for Claudin-Low Triple Negative Breast Cancer Reversion
Advisor: Dr. Paola Vera-Licona

Research Experience

Johns Hopkins Institute for Computational Medicine, Baltimore, MD — Aug. 2022 – present
Biomedical Engineering PhD Candidate, Advisors: Drs. Alison Hill and Feilim Mac Gabhann
Topic: Investigating within-host viral dynamics of persistent infections in response to novel cure strategies through mathematical models and statistical methods

- Building mechanistic mathematical models of persistent viral infections and evaluating their dynamic properties
- Developing Bayesian methods for parameter estimation and uncertainty quantification
- Designing stochastic and deterministic simulations to evaluate efficacy of therapeutic interventions for two persistent viral infections: Kaposi's sarcoma-associated herpesvirus (KSHV) and human immunodeficiency virus (HIV)
- Developing a pipeline for assessing practical identifiability of nonlinear mixed-effects models and reducing overfitting in data-driven covariate model building approaches
- Implementing nonlinear mixed-effects modeling to understand mechanisms of action of novel HIV immunotherapies
- Partnering with virologists to design in vitro studies using model-based predictions
- Outcomes: 2 publications (1 first author), 6 contributed talks, 4 poster presentations, 1 first author manuscript in preparation

Bristol Myers Squibb (BMS), Cambridge, MA — June 2026 – Aug. 2026
Clinical Pharmacology and Pharmacometrics Graduate Intern
Topic: Evaluating adaptive dosing of targeted cancer therapies to delay the onset of resistance

- Led the development of a tumor growth model to describe ecological interactions between drug-sensitive and drug-resistant subclones, calibrated to published data from a pilot clinical trial (mCRPC) via nonlinear mixed-effects modeling
- Designed and executed a virtual-population simulation to identify which tumor, drug, and dosing-schedule properties determine the clinical benefit of adaptive therapy, and determine the required tumor monitoring frequency for future trials
- Evaluated practical identifiability of model parameters from early clinical trial data to assess feasibility of informing dosing schedules in a prospective adaptive-therapy trial arm
- Outcomes: 1 internal talk to Clinical Pharmacology & Pharmacometrics teams, 1 company-wide internal poster presentation

Johns Hopkins Wilmer Eye Institute and Malone Center for Engineering in Healthcare, Aug. 2022 – June 2023
Baltimore, MD
Research Team Lead – Precision Care Medicine Course, Advisors: Drs. Alvin Liu, Craig Jones, Joseph Greenstein, and Casey Overby-Taylor
Topic: Developing an ensemble machine learning method to aid in clinical management of age-related macular degeneration

- Led a team of three undergraduates and one master's student in collaboration with clinical investigators as part of a full-year engineering course
- Oversaw the development of a multilayer perceptron neural network trained to predict progression to exudative age-related macular degeneration (eAMD) within 6 months, using both 3D optical coherence tomography images and clinical covariates associated with greater eAMD risk
- Evaluated the performance of our ensemble method against several classical machine learning methods trained on clinical covariates alone
- Outcomes: 1 publication, 2 internal talks, 1 internal poster presentation

The Jackson Laboratory (JAX), Bar Harbor, ME June 2020 – July 2022
Research Data Analyst, Advisor: Dr. Gary Churchill
Topic: A Bayesian approach to mediation analysis of complex traits with measurement noise

- Co-developed and validated an R package implementation of a Bayesian model selection approach to mediation analysis (bmediatR) that is flexible in both data inputs and potential inferences
- Diagnosed the effect of measurement noise on the inference of mediation using synthetic data and derived guidelines to avoid incorrect causal interpretation of spurious correlations in real-world genetic association examples
- Applied these tools to better understand the mechanisms underlying the effects of sex and diet on protein and gene expression in the livers of genetically diverse mice
- Outcomes: 2 publications (1 first author), 1 R package

UConn Health Center for Quantitative Medicine, Farmington, CT Sept. 2018 – May 2020
Undergraduate Research Assistant, Advisor: Dr. Paola Vera-Licona
Topic: A quantitative pipeline for cancer reversion analysis in triple negative breast cancer

- Constructed a static intracellular signaling network for a claudin-low triple negative breast cancer (CL TNBC) cell line with multi-omics data using bioinformatics techniques
- Applied a structure-based control method for nonlinear systems to identify putative targets that steer the system to any desired attractor
- Conducted virtual screenings using a network-based approach and a signal propagation algorithm to estimate long term behaviors
- Implemented machine learning methods to analyze virtual screenings and identify concerted perturbations of control nodes resulting in reversion of the CL TNBC phenotype
- Outcomes: 1 internal talk, 2 poster presentations, foundational pipeline for the Vera-Licona lab

Metrum Research Group, Tariffville, CT June 2018 – Aug. 2018
Summer Intern, Advisor: Dr. Ahmed Elmokadem
Topic: A maternal-fetal physiologically based pharmacokinetic model for drugs metabolized by cytochrome P450 isoenzymes

- Modeled maternal and fetal drug exposures at varying gestational ages by incorporating anatomical, biochemical, and physiological changes associated with pregnancy as a system of ordinary differential equations using R and *mrgsolve*
- Performed local sensitivity analysis, optimized model parameters, and validated the model by comparing model predictions to external published data
- Outcomes: 1 paper, 2 contributed talks, 1 poster presentation

University of Connecticut, Department of Molecular and Cellular Biology, Storrs, CT Jan. 2017 – Sept. 2017
Undergraduate Research Assistant/Holster Scholar, Advisor: Dr. Rachel O'Neill
Topic: The effect of host genetic variability on Epstein-Barr Virus (EBV)-associated cancer susceptibility

- Utilized wet lab techniques such as polymerase chain reactions, gel electrophoresis, and Sanger sequencing to sequence genes of interest in several EBV-associated cancer cell lines and a non-cancerous EBV+ control cell line
- Aligned the genetic sequence of target genes to identify common and distinct single nucleotide polymorphisms (SNPs) across EBV-associated cancer cell lines using the software Geneious and BLAST
- Outcomes: 1 internal talk

Manuscripts in Preparation

Gastonguay, M. S.[+], Gerold, J.[+], Balelli, I., Pasin, C., Li, J. Z., Barouch, D. H., Whitney, J. B., Hill, A. L.[+], & Prague, M.[+] (in preparation). Quantifying mechanisms of combination HIV immunotherapy using models of post-treatment rebound.
[+] *These authors contributed equally to the work*

Publications (ORCID: 0000-0002-5700-8543)

Juillard, F.[+], Gastonguay, M. S.[+], Szymula, A.[+], Raina, K., Tram, J., Gerold, J., Li, S., Lomonte, P., Simas, J. P., Hill, A. L., & Kaye, K. M. (in press). Maintenance of an episomal viral genome through imperfect replication and segregation mechanisms. *Nucleic Acids Research.*

Ferreira, I. A. T. M., Herrera, A., Huynh, T. T., *et al.,* Gastonguay, M. S., *et al.,* Jones, R. B. (2026). Dynamic antigen expression and cytotoxic T cell resistance in HIV reservoir clones. *Nature*, 653, 850–860. https://doi.org/10.1038/s41586-026-10298-w

Liu, T. Y. A., Liu, Y., Gastonguay, M. S., Midgett, D., Kuo, N., Zhao, Y., Ullah, K., Alexander, G., Hartman, T., Koseoglu, N. D., & Jones, C. (2025). Predicting imminent conversion to exudative age-related macular degeneration using multimodal data and ensemble machine learning. *Ophthalmology Science*. https://doi.org/10.1016/j.xops.2025.100785

Gastonguay, M. S., Keele, G. R., & Churchill, G. A. (2023). The trouble with triples: Examining the impact of measurement error in mediation analysis. *Genetics*, 224(1), iyad045. https://doi.org/10.1093/genetics/iyad045

Crouse, W. L., Keele, G. R., Gastonguay, M. S., Churchill, G. A., & Valdar, W. (2022). A Bayesian model selection approach to mediation analysis. *PLOS Genetics*, 18(5), e1010184. https://doi.org/10.1371/journal.pgen.1010184

Utsey, K., Gastonguay, M. S., Russell, S., Freling, R., Riggs, M. M., & Elmokadem, A. (2020). Quantification of the impact of partition coefficient prediction methods on physiologically based pharmacokinetic model output using a standardized tissue composition. *Drug Metabolism and Disposition*, 48(10), 903–916. https://doi.org/10.1124/dmd.120.090498

Zuppa, A. F., Benitez, G. R., Zane, N. R., Curley, M. A. Q., Bradfield, J., Hakonarson, H., Gastonguay, M. S., Moorthy, G., Prodell, J., & Gastonguay, M. R. (2019). Morphine Dose Optimization in Critically Ill Pediatric Patients With Acute Respiratory Failure: A Population Pharmacokinetic-Pharmacogenomic Study. *Critical Care Medicine*, *47*(6), e485–e494. https://doi.org/10.1097/CCM.0000000000003741

Zuppa, A. F., Conrado, D. J., Zane, N. R., Curley, M. A. Q., Bradfield, J., Hakonarson, H., Gastonguay, M. S., Moorthy, G., Prodell, J., & Gastonguay, M. R. (2019). Midazolam Dose Optimization in Critically Ill Pediatric Patients With Acute Respiratory Failure: A Population Pharmacokinetic-Pharmacogenomic Study. *Critical Care Medicine*, *47*(4), e301–e309. https://doi.org/10.1097/CCM.0000000000003638

[+]*These authors contributed equally to the work*

Oral Presentations

Quantifying mechanisms of HIV immunotherapy using models of post-treatment viral rebound, Invited talk for University of Pittsburgh Association for Women in Mathematics Seminar Series, Virtual, September 11th, 2026

Viral rebound kinetics following single and combination immunotherapy for HIV/SIV, Epidemics, San Diego, CA, December 1st, 2025

Quantifying the dynamics of Kaposi's sarcoma-associated herpesvirus persistence, Society for Mathematical Biology Annual Meeting, Edmonton, Canada, July 17th, 2025

Using mixed effects models to quantify the effects of single and combination immunotherapy on HIV/SIV rebound kinetics, Society for Mathematical Biology Annual Meeting, Edmonton, Canada, July 15th, 2025

Quantifying the dynamics of Kaposi's sarcoma-associated herpesvirus persistence, Dynamics & Evolution of Human Viruses, Paris, France, May 9th, 2025

Quantifying the dynamics of Kaposi's sarcoma-associated herpesvirus persistence, Epidemics, Bologna, Italy, November 27th, 2023

Predicting imminent conversion to exudative age-related macular degeneration using multimodal data and ensemble machine learning, Johns Hopkins Engineering Design Day, Baltimore, MD, May 2nd, 2023

Prediction of maternal-fetal exposures of CYP450-metabolized drugs using physiologic pharmacokinetic modeling implemented in R and mrgsolve, R/pharma Conference, Cambridge, MA, August 23rd, 2019

Identification of Combinations of Targets for Claudin-Low Triple Negative Breast Cancer Reversion, UConn Center for Cell Analysis and Modeling Summer Seminar, Farmington, CT, July 26th, 2019

Development of an Open and General Physiologically Based Pharmacokinetic Model to Predict Maternal-Fetal Exposures for Drugs Metabolized by CYP Isoenzymes, R/Medicine Conference, New Haven, CT, September 8th, 2018

The Effect of Host Genetic Variability on Epstein-Barr Virus-associated cancer susceptibility, UConn Holster Scholar Symposium, Storrs, CT, October 2017

Poster Presentations

Modeling viral rebound kinetics following single and combination immunotherapy for human immunodeficiency virus, American Conference on Pharmacometrics, National Harbor, MD, October 11-14th, 2026

Treatment to Contain: how model-informed adaptive cancer therapy can delay disease progression, Bristol Myers Squibb Intern Poster Session (internal), Cambridge, MA, August 4th, 2026

Viral rebound kinetics following single and combination immunotherapy for HIV/SIV, Dynamics & Evolution of Human Viruses, Kelowna, Canada, May 21st, 2026

Quantifying the dynamics of Kaposi's sarcoma-associated herpesvirus persistence, MIDAS Network Annual Meeting, Bethesda, MD, November 18th, 2024

Quantifying the dynamics of Kaposi's sarcoma-associated herpesvirus persistence, International Workshop on Kaposi's Sarcoma Herpesvirus & Related Agents, Boston, MA, July 7th, 2024

Predicting imminent conversion to exudative age-related macular degeneration using multimodal data and ensemble machine learning, Johns Hopkins Engineering Design Day, Baltimore, MD, May 2nd, 2023

Identification of Combinations of Pharmacologic Targets for Claudin-Low Triple Negative Breast Cancer Reversion, International Society of Pharmacometrics Quantitative Systems Pharmacology Student Symposium, Virtual, April 28th, 2021

Identification of Combinations of Targets for Claudin-Low Triple Negative Breast Cancer Reversion, Joint Meeting in Mathematics, Denver, CO, January 15th – 18th, 2020

Development of an Open-source Physiologically-Based Pharmacokinetic Model to Predict Maternal-Fetal Exposures of CYP450-Metabolized Drugs, International Society of Pharmacometrics Regional Quantitative Systems Pharmacology Day, Princeton, NJ, July 16th, 2019

Open-Source Contributions

bmediatR, an open-source R package implementing a Bayesian model selection approach to mediation analysis. May 2022
Available to install from GitHub https://github.com/wesleycrouse/bmediatR

- Contributed to methods development and validation in both synthetic and real data
- Designed and built package vignettes

Fellowships

Summer Undergraduate Research Fund, University of Connecticut Office of Undergraduate Research May 2019

- Awarded funding for a 9-week summer research project

Holster Scholar, University of Connecticut Honors Program May 2017

- Offered enrollment in a semester-long course to learn how to develop and write a project proposal
- Selected as one of 8 students awarded funding for a 10-week summer research project

Honors and Awards

Blue Ribbon Poster Award, ISoP Quantitative Systems Pharmacology Student Symposium April 2021
Dean's List, The University of Connecticut Sept. 2016 – May 2020
Academic Excellence Scholarship, The University of Connecticut Sept. 2016 – May 2020
Babbidge Scholar, The University of Connecticut Dec. 2017, 2019
New England Scholar, The University of Connecticut Dec. 2018

Teaching Experience

Dynamic modeling of infectious diseases in patients and populations, Johns Hopkins University (TA) Oct. – Dec. 2024
Systems pharmacology and personalized medicine, Johns Hopkins University (TA) Feb. – May 2024
Data Carpentry ecology with R workshop, Bioinformatics Training Program at JAX (Instructor) Dec. 2021
Data Carpentry genomics workshop, Bioinformatics Training Program at JAX (Instructor) Nov. 2021
Introductory statistics with R, Bioinformatics Training Program at JAX (TA) Sept. 2021
Introduction to R and RStudio, Bioinformatics Training Program at JAX (TA) June 2021

Professional Development

International Society of Pharmacometrics Member 2025 – Present
Society for Mathematical Biology Member 2025 – Present
Hybrid Modeling Strategies for Drug Development and Decision-Making Tutorial, Oct. 2026
American Conference on Pharmacometrics
Summer Institute in Statistics and Modeling in Infectious Diseases, Emory University July 2024

Simulation Modeling in Immunology by Andreas Handel and Paul Thomas	Feb. 2023
Building Tidy R Packages, R/pharma 2021	Oct. 2021
Julia Language for R Programmers, R/pharma 2021	Oct. 2021
Advanced Shiny, RStudio	Sept. 2021
Introduction to Bayesian Data Analysis, JuliaCon	July 2021
Carpentries Instructor Training, The Carpentries	Mar. 2021
Shiny, RMarkdown, and RStudio Connect, RStudio	Mar. 2021
Quantitative Trait Mapping in the Diversity Outbred, University of Wisconsin-Madison	Dec. 2020
Containerization with Singularity, JAX	Oct. 2020
Introduction to HPC, JAX	Sept. 2020
Human and Mammalian Genetics and Genomics: The 61st McKusick Short Course, JAX	July 2020
Shiny Reproducibility, R/pharma 2019	Aug. 2019
Machine Learning, R/pharma 2019	Aug. 2019

Service & Involvement

Student Mentor, Johns Hopkins University Women Mentoring Women Program	2025 – Present
Member of Synaptic Clefs A Cappella, Johns Hopkins School of Medicine	2023 – Present
BME Application Assistance Program, Johns Hopkins University	2022 – Present
Math Motivators High School Tutoring, The University of Connecticut	2017 – 2019
iGEM Genetic Engineering Team, The University of Connecticut	2016 – 2017
President & Treasurer of Rubyfruit A Cappella, The University of Connecticut	2016 – 2020